Statistics & Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

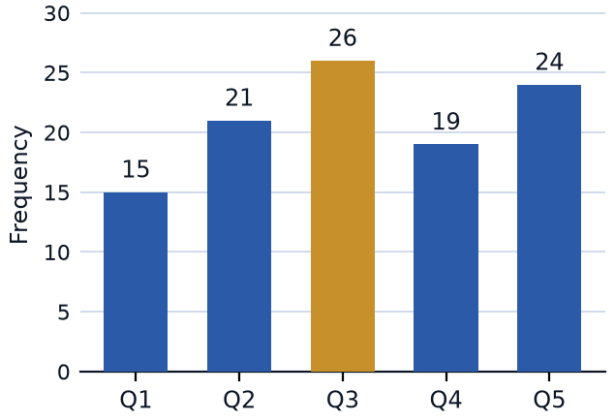
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the range of the data set: 20, 39, 10, 12, 26.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the range of the data set: 38, 47, 25, 57, 46.



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the range of the data set: 91, 86, 36, 84, 15, 44, 11.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

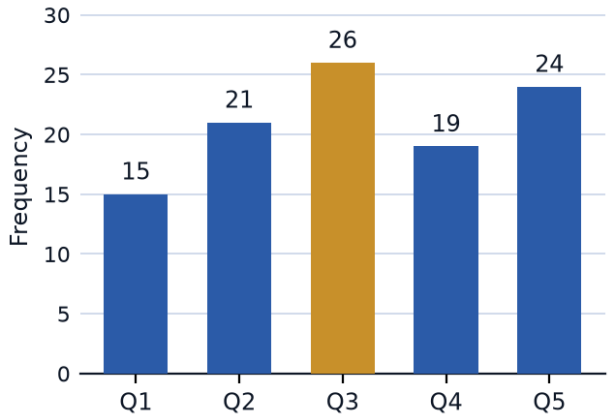
Find the range of the data set: 97, 12, 10, 30, 50, 81, 49.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the range of the data set: 87, 89, 26, 14, 61, 66, 77, 99, 38.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the range of the data set: 41, 97, 11, 85, 63, 39, 69.



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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the range of the data set: 8, 23, 19, 12, 3, 11.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

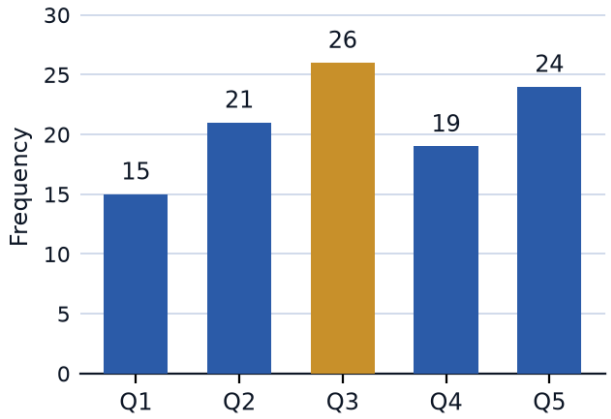
Find the range of the data set: 25, 9, 11, 28, 18.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the range of the data set: 16, 5, 15, 21, 20, 15.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the range of the data set: 8, 27, 20, 25, 20.



11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the range of the data set: 15, 20, 16, 53, 9.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the range of the data set: 58, 31, 1, 33, 54.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the range of the data set: 20, 42, 86, 77, 5, 81, 76, 99, 72.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the range of the data set: 66, 79, 70, 57, 8, 48, 40, 2, 14.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

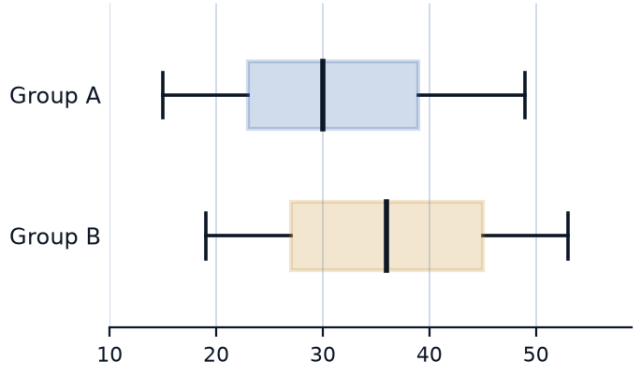
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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

Find the median of the data set: 32, 10, 29, 13, 37, 36, 34, 38.

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

Find the median of the data set: 24, 11, 20, 9, 34, 15, 43, 29.



17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Find the median of the data set: 11, 38, 64, 14, 28, 71, 8, 43.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Functions & Graphs

Find the median of the data set: 16, 6, 55, 25, 71, 72, 77, 74.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Coordinate Geometry

Two independent events A and B have $P(A) = \frac{3}{4}$ and $P(B) = \frac{9}{10}$. Find $P(A \text{ and } B)$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Plane Geometry

Two independent events A and B have $P(A) = \frac{1}{5}$ and $P(B) = \frac{3}{10}$. Find $P(A \text{ and } B)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A student scored $82, 88, 91,$ and 85 on four exams. What score must the student earn on the fifth exam to achieve an overall mean score of 88?

a. 90

b. 92

c. 94

d. 96

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a set of 7 integers: $\{4, 7, 9, x, 14, 18, 20\}$ arranged in ascending order, the median is equal to the mean. What is the value of x?

a. 12

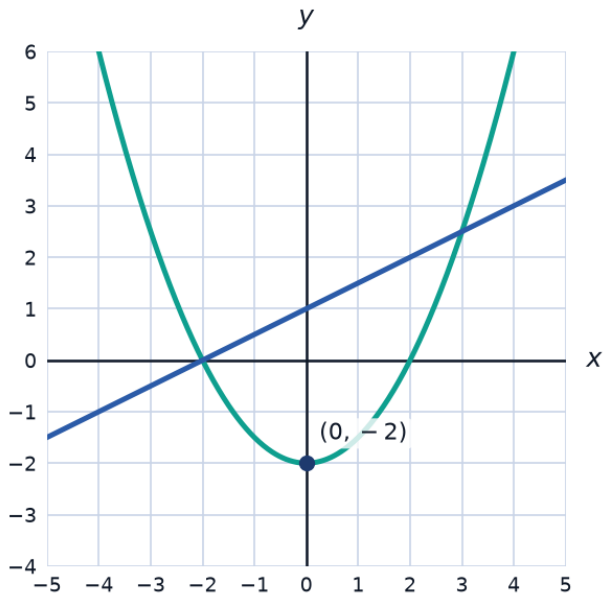
b. 13

c. 14

d. 15

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a chemistry course, homework counts for 20%, labs count for 30%, and tests count for 50%. If a student earns 90 on homework, 80 on labs, and 86 on tests, what is the weighted final grade?



a. 84

b. 85

c. 86

d. 87

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A data set consists of the numbers $\{3, 7, 8, 12, 15, 18, 21\}$. What is the interquartile range (IQR) of this data set?

a. 9

b. 11

c. 12

d. 14

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If every value in a data set is multiplied by 3 and then increased by 5, how is the standard deviation σ affected?

a. It increases by 5

b. It is multiplied by 3

c. It is multiplied by 3 and increased by 5

d. It remains unchanged

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which of the following data sets has a standard deviation equal to 0?

a. $\{5, 5, 5, 5, 5\}$

b. $\{1, 2, 3, 4, 5\}$

c. $\{-2, 0, 2\}$

d. $\{10, 20, 30\}$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A bag contains 6 red marbles, 8 blue marbles, and 10 green marbles. If one marble is selected at random, what is the probability that it is NOT blue?

a. $\frac{1}{3}$

b. $\frac{2}{3}$

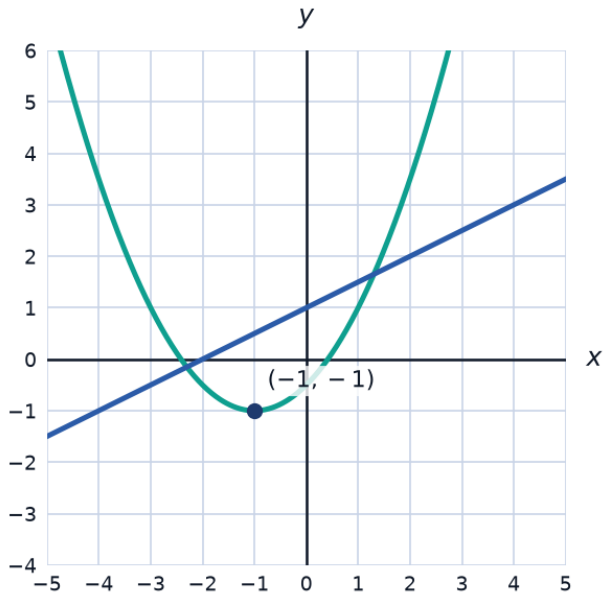
c. $\frac{1}{4}$

d. $\frac{3}{4}$

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two fair standard six-sided dice are rolled simultaneously. What is the probability that the sum of the two numbers rolled is greater than 9?



a. $\frac{1}{6}$

b. $\frac{5}{36}$

c. $\frac{1}{9}$

d. $\frac{7}{36}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Events A and B are independent. If $P(A) = 0.4$ and $P(B) = 0.5$, what is the probability that at least one of the events occurs, $P(A \cup B)$?

a. 0.70

b. 0.80

c. 0.90

d. 0.20

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In how many distinct ways can a committee of 4 members be chosen from a group of 9 people?

a. 36

b. 84

c. 126

d. 3024

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How many different 4-digit identification codes can be formed using digits 0 through 9 if repetition of digits is not allowed and the first digit cannot be 0?

a. $4,536$

b. $5,040$

c. $9,000$

d. $3,024$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A restaurant offers 4 appetizers, 6 main courses, and 3 desserts. How many distinct 3-course dinner combinations are possible?

a. 13

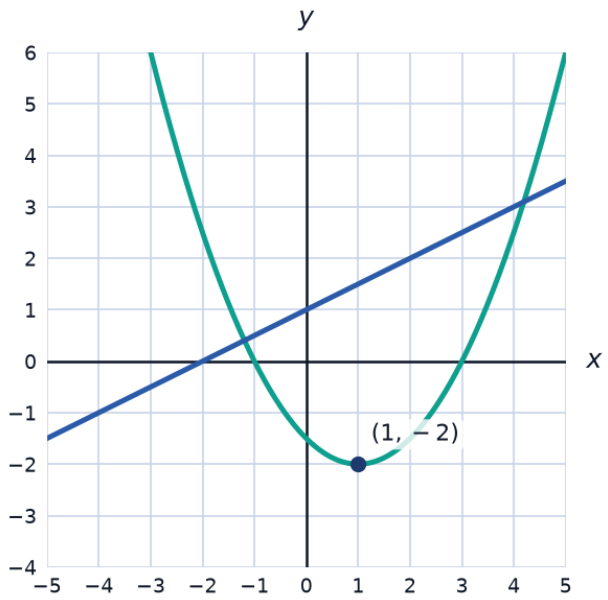
b. 36

c. 72

d. 144

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two independent events A and B have $P(A) = \frac{3}{4}$ and $P(B) = \frac{1}{5}$. Find $P(A \text{ and } B)$.



a. $\frac{3}{50}$

b. $\frac{3}{5}$

c. $\frac{3}{4}$

d. $\frac{3}{20}$

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34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two independent events A and B have $P(A) = \frac{1}{6}$ and $P(B) = \frac{1}{2}$. Find $P(A \text{ and } B)$.

a. $\frac{4}{5}$

b. $\frac{3}{5}$

c. $\frac{1}{20}$

d. $\frac{1}{12}$

35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two independent events A and B have $P(A) = \frac{5}{6}$ and $P(B) = \frac{9}{10}$. Find $P(A \text{ and } B)$.

a. $\frac{1}{2}$

b. $\frac{4}{5}$

c. $\frac{5}{8}$

d. $\frac{3}{4}$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two independent events A and B have $P(A) = \frac{2}{3}$ and $P(B) = \frac{3}{5}$. Find $P(A \text{ and } B)$.

a. $\frac{1}{2}$

b. $\frac{2}{5}$

c. $\frac{1}{4}$

d. 0

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two independent events A and B have $P(A) = \frac{1}{4}$ and $P(B) = \frac{2}{3}$. Find $P(A \text{ and } B)$.

a. $\frac{2}{3}$

b. $\frac{1}{6}$

c. 0

d. $\frac{3}{4}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Functions & Graphs

Two independent events A and B have $P(A) = \frac{5}{6}$ and $P(B) = \frac{2}{5}$. Find $P(A \text{ and } B)$.

a. $\frac{1}{2}$

b. $\frac{1}{8}$

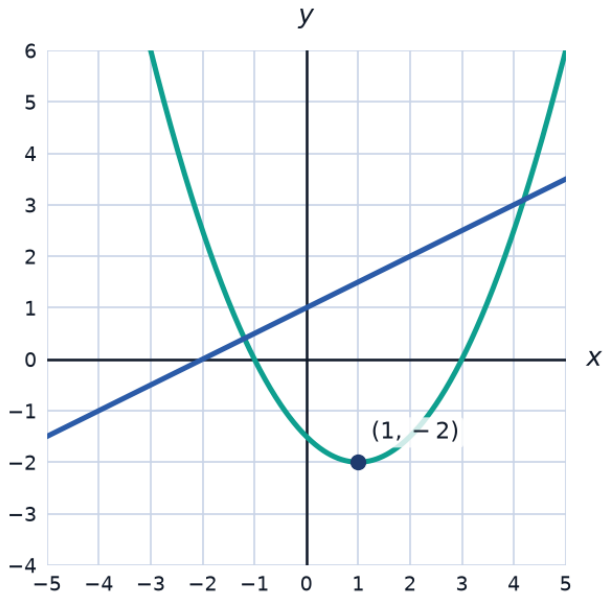
c. $\frac{3}{10}$

d. $\frac{1}{3}$

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39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

Two independent events A and B have $P(A) = \frac{5}{6}$ and $P(B) = \frac{1}{5}$. Find $P(A \text{ and } B)$.



a. $\frac{2}{5}$

b. $\frac{1}{6}$

c. $\frac{16}{25}$

d. 1

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Plane Geometry

Two independent events A and B have $P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{4}$. Find $P(A \text{ and } B)$.

a. 0

b. $\frac{1}{8}$

c. $\frac{1}{3}$

d. 1

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

Find the median of the data set: 35, 22, 1, 6, 3, 49.

a. 35

b. 8

c. 29

d. 14

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

A data set of 9 values has a mean of 54. A new value of 224 is added. What is the new mean?

a. 43

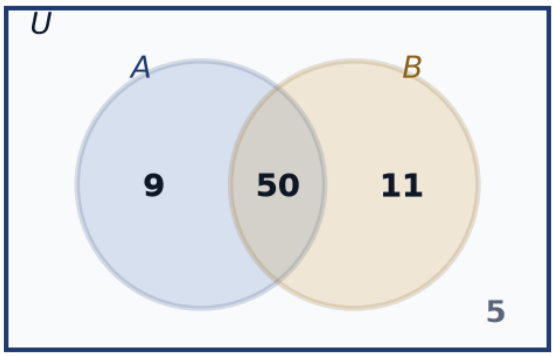
b. 71

c. 47

d. 138

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

A data set of 9 values has a mean of 50. A new value of 340 is added. What is the new mean?



a. 140

b. 79

c. 4

d. 164

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

A data set of 9 values has a mean of 29. A new value of 179 is added. What is the new mean?

a. 69

b. 44

c. 21

d. 39

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Functions & Graphs

A data set of 7 values has a mean of 33. A new value of 113 is added. What is the new mean?

a. 82

b. 43

c. 97

d. 39

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